

Observational techniques

Gather information about actual use of a system **by observing users interacting** with it.

Techniques:

- **Thinking aloud**
- **Cooperative evaluation**

Related topics:

- **Protocol analysis** (methods for recording user actions)
- **Post-task walkthrough** (to understand why actions were performed)



1

Thinking aloud (observational technique)

- The user is asked to **talk through what he is doing** as he is being observed..
- ..describing **what he believes** is happening, **why** he takes an action, what he is trying to do.

and now..
I look for the shopping cart..



2

Thinking aloud

Advantages:

- **simplicity** (requires little expertise)
- can provide **useful insight** into problems
- can show how system is **actually used**

Limitations:

- **subjective** (captures what users think they are doing, not necessarily what they are actually doing)
- **selective** (reveals the **problems associated with the tasks selected** for the evaluation)
- act of **describing may alter** task performance



3

Cooperative evaluation (observational technique)

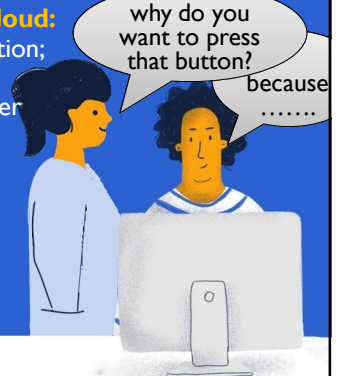
Variation on thinking aloud:

user collaborates in evaluation; both user and (**expert**) **evaluator** can ask each other questions throughout

why do you want to press that button?
because

Additional advantages:

- less constrained and easier to use
- user is encouraged to criticize system
- clarification possible



4

Protocol analysis

The **record of an evaluation session** is known as a **protocol**. Several methods:

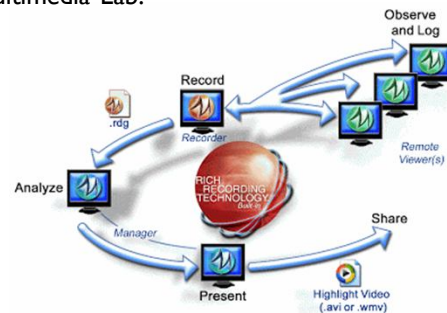
- **paper and pencil**: cheap, limited to writing speed
- **audio**: good for think aloud, difficult to match with other protocols
- **video**: accurate and realistic, needs special equipment, obtrusive
- **computer logging**: automatic and unobtrusive, large amounts of data difficult to analyze, longitudinal studies
- user **notebooks** (by the participants themselves): coarse and subjective, useful insights, good for longitudinal studies



5

Usability testing with Morae (discontinued) 1/2

Used for some time in the Computer Vision and Multimedia Lab.



6

Post-task walkthrough

Direct observation methods (including think-aloud) may be **incomplete**. (Why certain actions have been performed? What about alternative actions?)

During a **post-task walkthrough**, immediately after the experiment, the transcript (or the video) is replayed to the participant who is invited to comment or to answer the questions by the analyst.



7

Post-task walkthrough

Advantages:

- may be the only way to obtain a subjective viewpoint
- analyst has time to focus on relevant incidents
- avoid excessive interruption of task

Limitations:

- lack of freshness
- may be **post-hoc interpretation** of events

Since **that** event followed **this** one, **that** event must have been caused by **this** one...



8

Query techniques

Asking the user about the interface directly may help to understand how to design a better interface.

Query techniques can be used in **evaluation** and more widely to collect information about user **requirements** and tasks

Two main types:

- **interviews**
- **questionnaires**



9

Interviews

Analyst questions user on **one-to-one** basis, usually starting from a **set of central questions** (planned in advance)

Top-down approach (from general questions to details about certain aspects)

Informal, subjective and relatively cheap

Can be effective for **high-level evaluation** (user preferences..)



10

Interviews

Advantages:

- **flexibility**: can be varied to suit context
- issues can be explored more fully (when needed)
- can elicit user views and identify **unanticipated problems**

Limitations:

- very **subjective**
- **difficult to standardize**;
- **time consuming**



11

Questionnaires

Set of fixed questions given to users

Advantages:

- **quick** and reaches **large user group**
- can be **analyzed** more **rigorously**
- **less intrusive** than interviews

Limitations:

- less flexible
- no follow-up questions
- unexpected issues may remain unexplored



12

Questionnaire design

Questionnaires need careful design:

- Which is the purpose? What information is required?
- How are answers to be **analyzed**?
(e.g. Do we need to calculate averages..?)

Styles of questions:

- **general** (background: age, sex, occupation..)
- **open-ended** (opinion, subjective, difficult to analyze)
- **scalar** (*It is easy to recover. Disagree | 2 3 4 5 Agree*)
- **multi-choice** (explicit responses)
- **ranked** (ordering items within a list; user's preferences)



13

Drag & Drop Ranking

Please rank the following ice cream flavors in order from most to least favorite.
Drag items from the left-hand list into the right-hand list to order them.

Chocolate	→	
Vanilla	→	
Strawberry	→	
Rocky Road	→	
Mint Chocolate Chip	→	
Cookies and Cream	→	

14

Ranking Grid

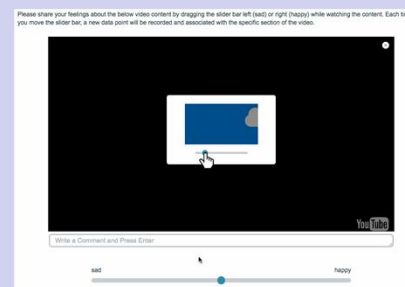
Please rank the following ice cream flavors.

	1	2	3	4	5	6
Chocolate	☐	☐	☐	☒	☐	☐
Vanilla	☐	☐	☐	☐	☒	☐
Strawberry	☐	☒	☐	☐	☐	☐
Rocky Road	☐	☐	☒	☐	☐	☐
Mint Chocolate Chip	☐	☐	☐	☐	☐	☒
Cookies and Cream	☒	☐	☐	☐	☐	☐

15

Video sentiment question

The **Video Sentiment** question type allows your survey respondents to provide continuous feedback while watching a video, by dragging a slider along a range



16

Designing Effective Questionnaires

Closed questions (scalar, ranked or multi-choice) **reduce the effort** required, ensure a **high response rate** and are easier to analyze.

Analysis may include search of **correlations**.

Carry out a **pilot study** with few users!

Questionnaires are now often distributed via the **internet**... but... questionnaire respondents are **self-selecting** anyway, in that only those who choose to respond are included in the study (they are not representative).



17

Objective usability testing

Evaluation through **monitoring physiological responses**.

Measuring.. instead of asking the users..

Two main areas:

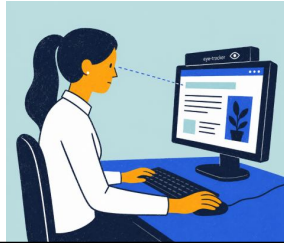
- **eye-tracking**
- **other physiological measurements**



18

Eye tracking

- head or desk mounted equipment tracks the position of the eye
- eye movement reflects the amount of cognitive processing a display requires
- combined with other techniques such as observation, interviews, or Thinking Aloud



19

Eye tracking measures

Measurements include

- **Fixations:** eye maintains stable position. Number and duration indicate level of difficulty with display
- **Saccades:** rapid eye movement from one point of interest to another
- **Scan paths:** moving straight to a target with a short fixation at the target is optimal



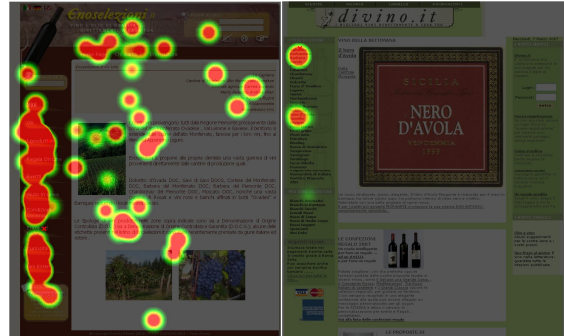
20

Gaze plots (single user)



21

Heat maps (one or many users)



22

Physiological measures

Emotional responses are linked to **physical changes** (heart rate, breathing and skin secretions...)

These may help determine a user's reaction to an interface (e.g. which **interaction events** really cause a user stress?)

Physiological measures are often **combined** with observation, interviews, or eye tracking



23

Physiological measures examples

Heart activity, indicated by blood pressure, volume and pulse. These may respond to **stress** or **anger**.

Activity of the sweat glands, indicated by skin resistance or galvanic skin response (GSR). These are thought to indicate levels of **arousal** and **mental effort**.

Electrical activity in muscle, measured by the electromyogram (EMG). These appear to reflect **involvement in a task**.

Electrical activity in the brain, measured by the electroencephalogram (EEG). These are associated with **decision making, attention and motivation**.

24